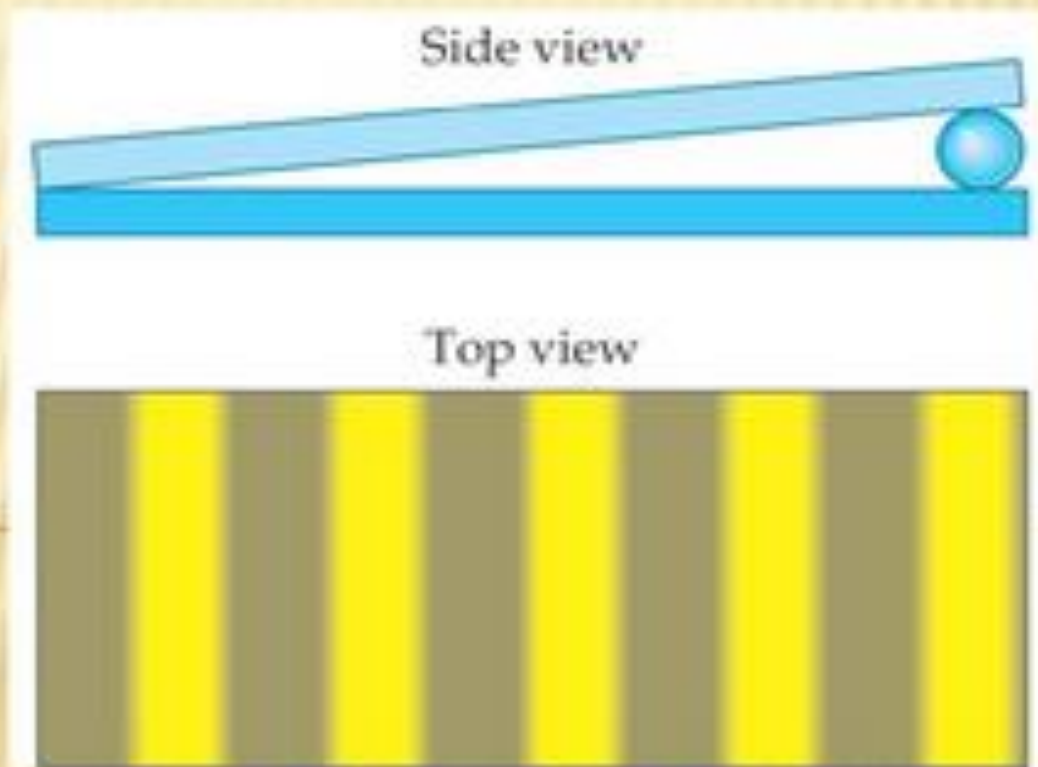
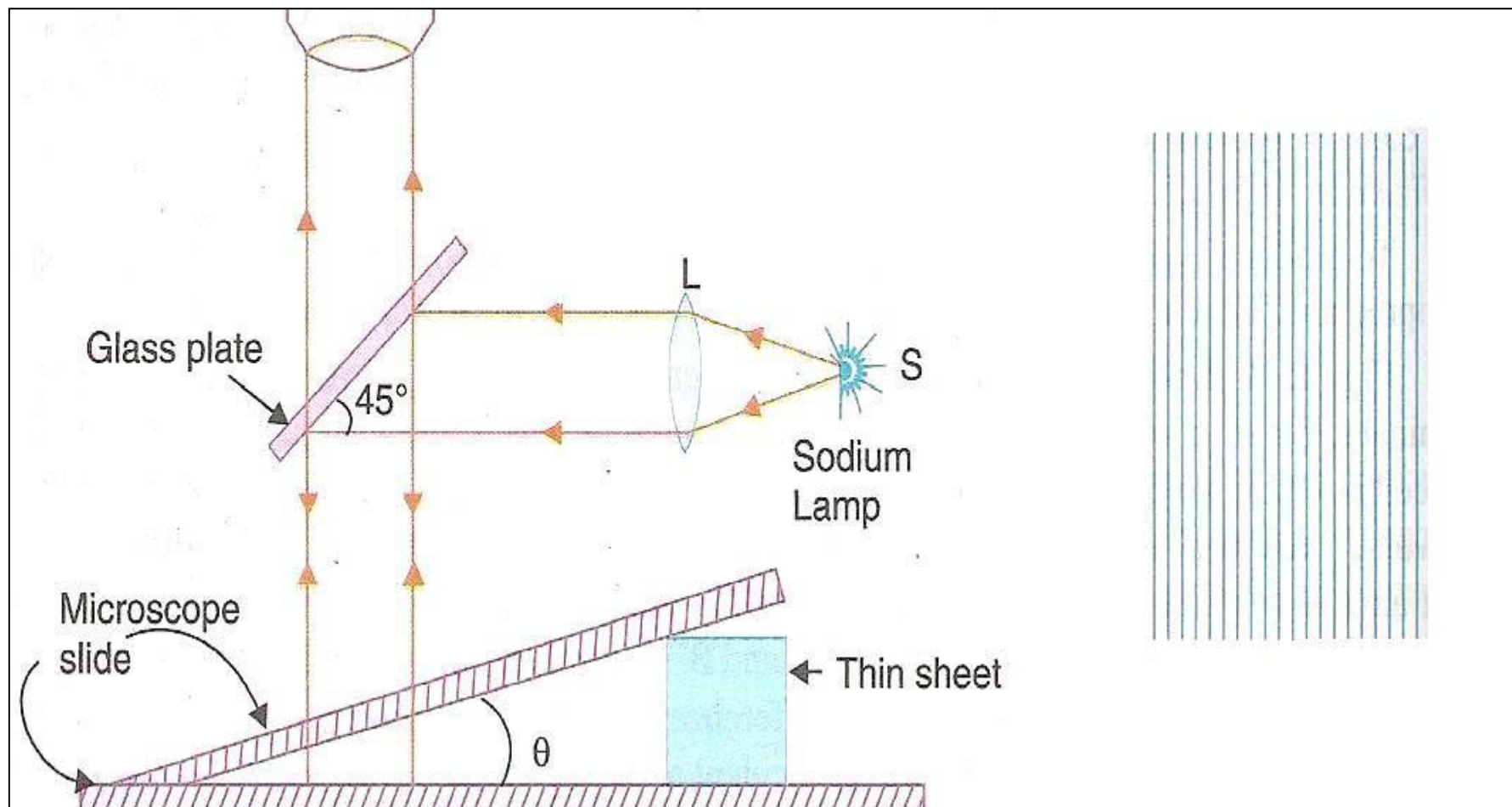


WEDGE SHAPED FILM

Variable Thickness (Wedge shaped) films

- Film with zero thickness at one end, progressively increasing to a particular thickness at the other end.
- Wedge angle very small, fraction of a degree.



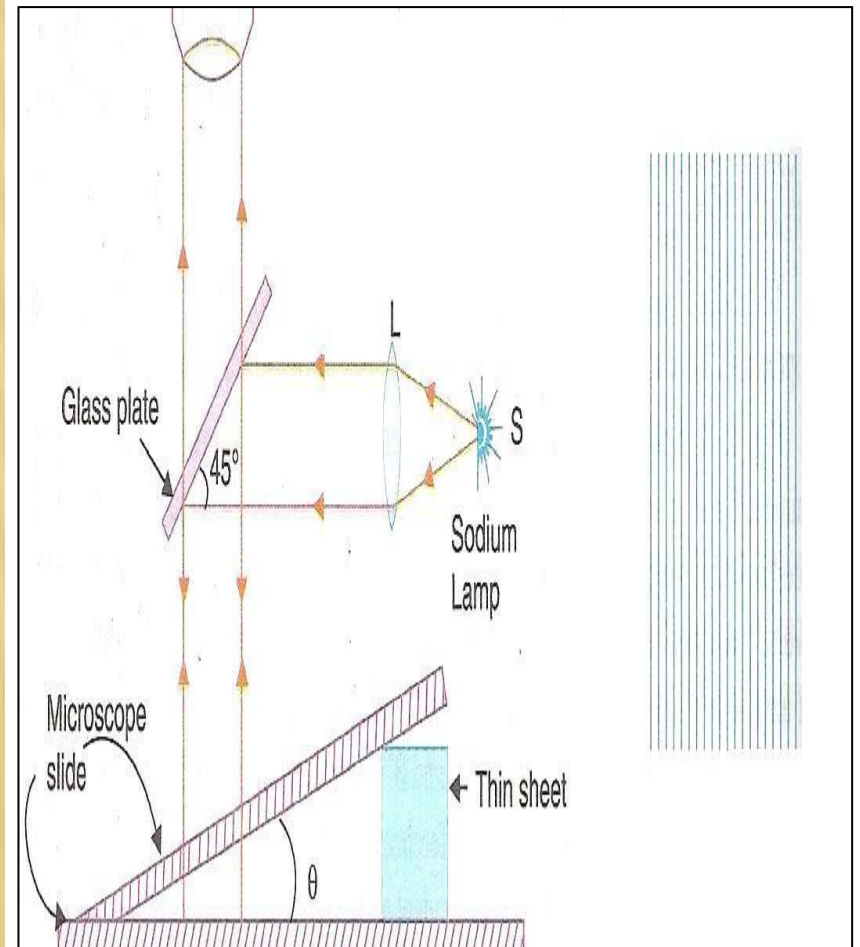


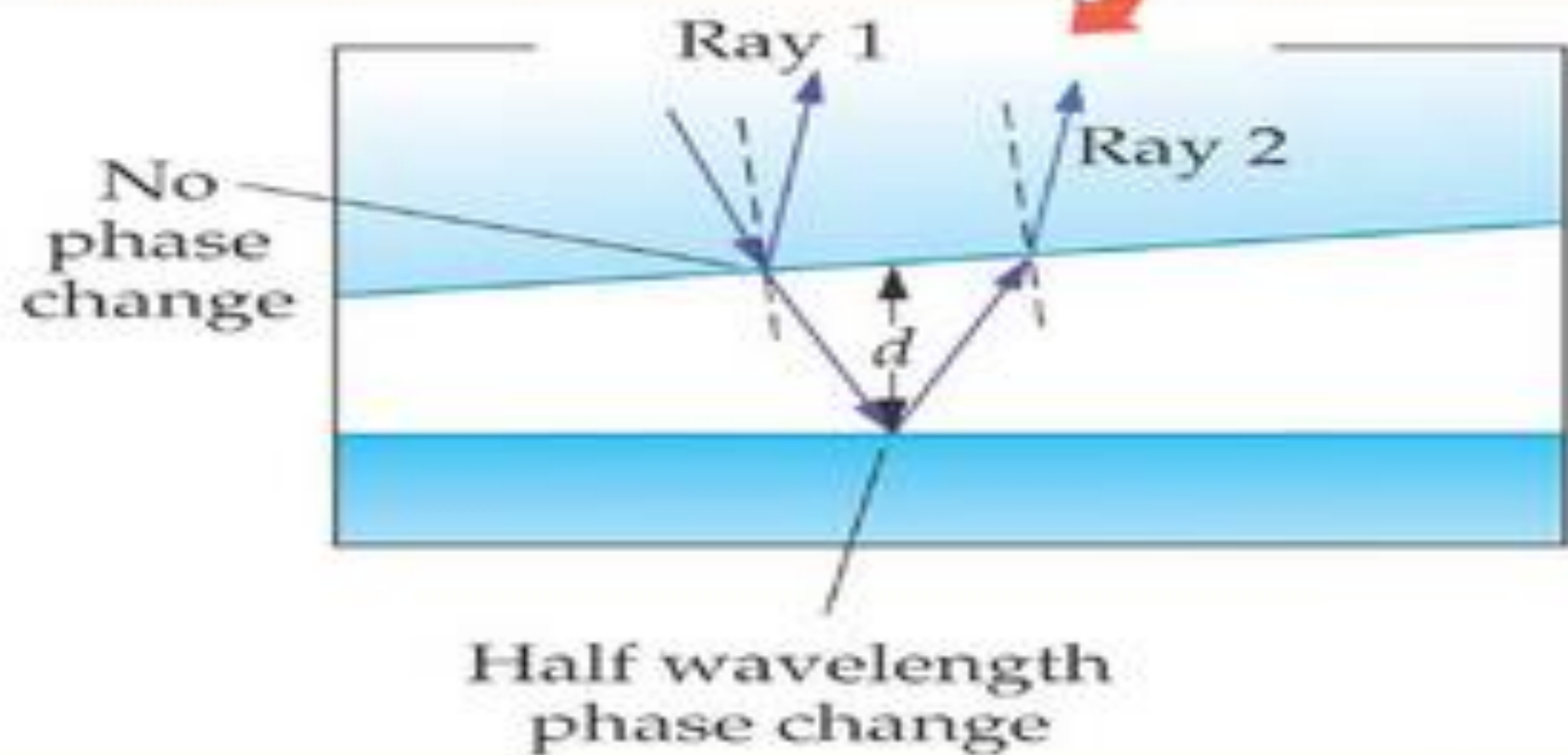
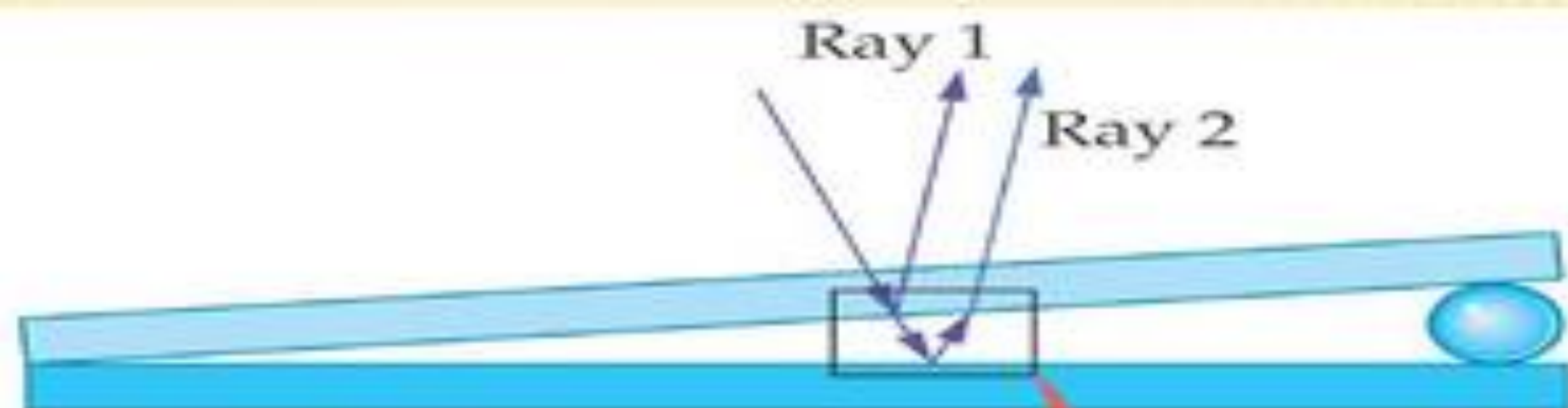
➤ When a parallel beam of monochromatic light illuminates the wedge from above

➤ the rays reflected from the two bounding surfaces will not be parallel and diverge from a point near the film.

➤ Path difference between the reflected rays from the lower and upper surfaces of the air film varies along its length due to variation in film thickness.

➤ Alternate bright and dark fringes on its top surface.

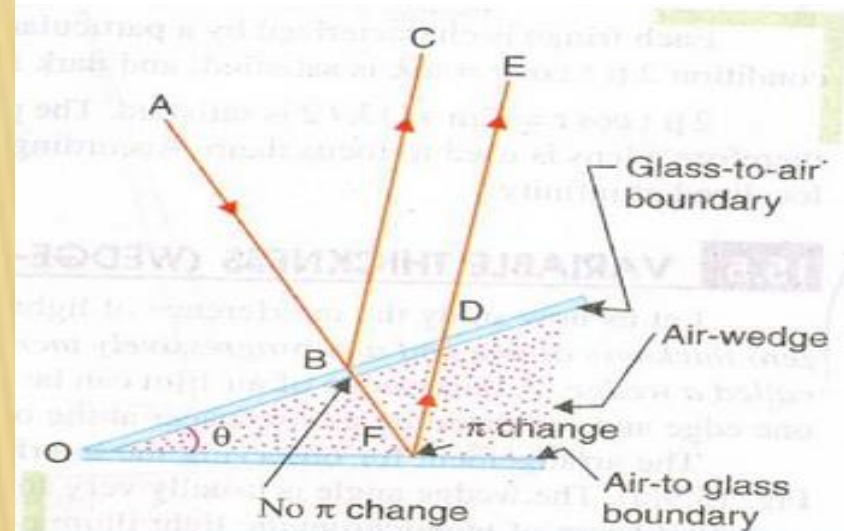




➤ BC and DE reflected from the top and bottom of the air film are coherent, derived from the same ray AB (division of amplitude)

➤ The rays are very close if the thickness of the film is of the order of wavelength of light.

➤ Interference of rays --- fringes depending on the phase difference for small air film thickness



➤ Thickness of the glass plate large as compared to the wavelength of incident light.

➤ Optical difference between BC and DE.

$$\Delta = 2\mu t \cos r - \frac{\lambda}{2}$$

➤ where $\frac{\lambda}{2}$ is the abrupt phase change of on reflection from the boundary of air to glass interface.
(Point F)

➤ For Maxima

$$\Delta = m\lambda$$

➤ Path difference

$$\Delta = 2\mu t \cos r - \frac{\lambda}{2}$$

➤ For bright fringe

$$2\mu t \cos r = (2m + 1)\frac{\lambda}{2}$$

➤ For minima

$$\Delta = (2m + 1)\frac{\lambda}{2}$$

➤ For dark fringe

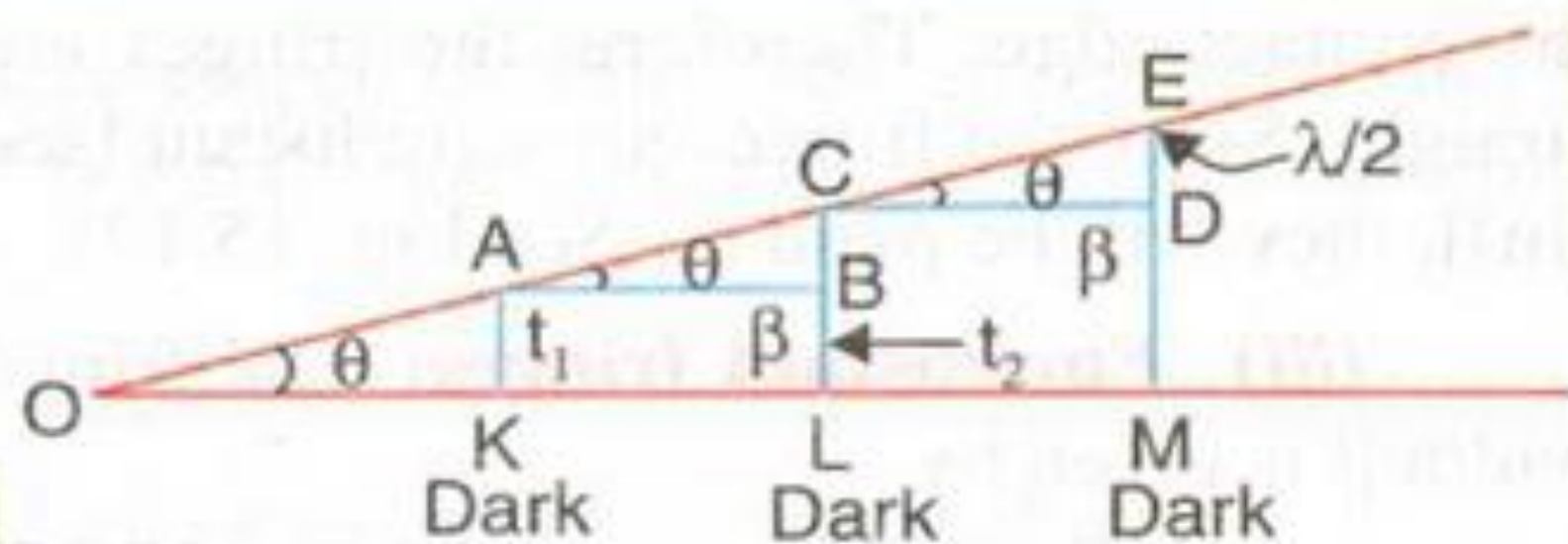
$$2\mu t \cos r = m\lambda$$

Assuming normal incidence $\cos r = 1$,
thickness of film at A = t_1

then at A, a dark fringe is formed (say)

$$2\mu t_1 = m\lambda \quad (1)$$

Next dark fringe occurs at C where, the thickness
 $CL = t_2$, then at C, $2\mu t_2 = (m+1)\lambda \quad (2)$



✖ Subtracting (1) from (2)

$$2\mu(t_2 - t_1) = \lambda$$

But, $(t_2 - t_1) = BC$

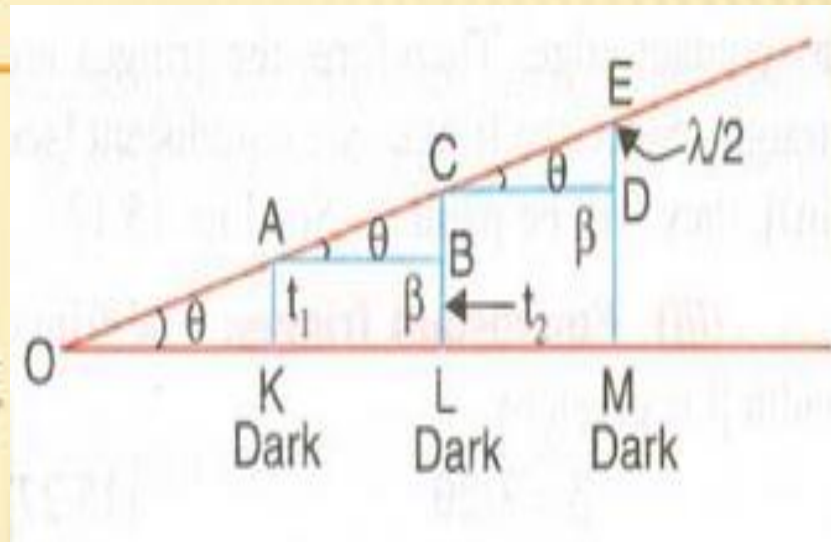
Therefore, $2\mu(BC) = \lambda$

$$BC = \frac{\lambda}{2\mu}$$

From $\triangle ABC$, angle $CAB = \theta$

and $BC = AB \tan \theta$

- ✗ AB is the distance between successive dark fringes (bright fringes also) $\rightarrow$ fringe width (β)





Then

$$\beta = \frac{\lambda}{2\mu \tan \theta}$$

For small values of θ , $\tan \theta \approx \theta$

Therefore,

$$\beta = \frac{\lambda}{2\mu \theta}$$

From the above equation fringe width β is constant for given wedge angle.

As θ increases, β decreases. (fringes move closer).

As the value of θ , approaches 90° , the interference pattern vanishes.

The interference pattern has the following salient features:



1. Fringe at the apex is dark,
2. Fringes are straight and parallel,
3. Fringes are equidistant,
4. Fringes are localized,
5. Fringes are of equal thickness



➤ Fringe at Apex is Dark

- At Apex, the two glass slides are in contact with each other.
- Therefore, the thickness $t \approx 0$
- The optical path difference then becomes

$$\Delta = 2\mu t - \frac{\lambda}{2} = 0 - \frac{\lambda}{2} = -\frac{\lambda}{2}$$

- It implies that the path difference of $\frac{\lambda}{2}$ or a phase difference of π occurs between the reflected waves at the edge.
- The two waves interfere destructively.
- Hence the fringe at the apex is dark



➤ **Straight and parallel fringes**

➤ Each fringe pattern produced by the interference of rays reflected from the section of the wedge have same thickness.

Locus of points having same thickness lie along lines parallel to the contact edge → **Straight and parallel edges**

➤ **Equidistant fringes**

$$\beta \approx \frac{\lambda}{2\theta}$$

since λ and θ are constants, β is a constant for a given wedge angle (hence equidistant fringes)

➤ **Localized fringes**

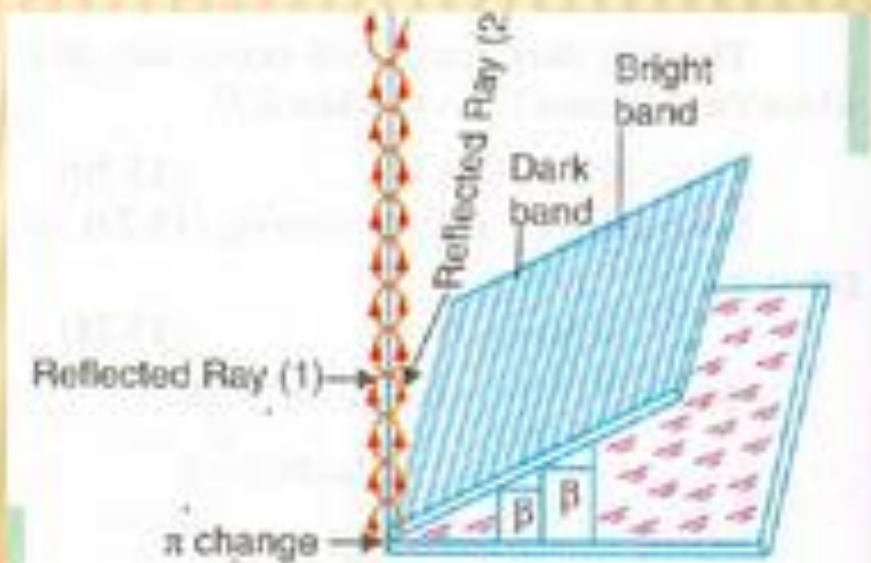
Fringes formed close to the top surface of wedge.

➤ Fringes of equal thickness

For films of thickness $\approx$ few λ , the rays from various parts of the film have same inclination (app.)

Δ between the overlapping waves changes mainly due to the change in thickness $\rightarrow$ **Fringes due to variation of film thickness**

Each fringe will be the locus of points of same thickness. Hence called the **fringes of equal thickness**.



THANK YOU